Rust Programming Language

A high level overview of the latest highly admired programming language by developers.

Feedback

Authors reread their sentences are they write them.

Looking for a yellow car - they are out there but you won't notice them until you start looking.

Why Rust?

Blazing Fast

"Don't count your servers, make your servers count."

- CloudFlare open sources their http proxy (alternative t
- The White House Office of the National Cyber Director
  risk of cyberattacks by using programming languages t
  vulnerabilities.
- [NSA cybersecurity information sheet] recommends we
  (https://media.defense.gov/2022/Nov/10/2003112742/-
  C#, Go, Java, Ruby, Swift and Rust.
- Rust developers at Google are twice as productive as C

> "So we see reduced memory usage in the services that w
> decreased defect rate over time in those services that h
> correctness."

- Zed.dev
- Project Oxidation at Mozilla[https://wiki.mozilla.org/Ox

Why not Rust?

- Limits the creative flow with it's constraints. No driving your car off the cliff for fun.
- Compiler is your friend and your foe. Lifetimes are difficult and the compiler will give you hints.
- Large cognitive load to learn.
- Compile time is longer (than Go, etc)
- You want quick code to production and you are happy to have errors in production - Rust causes you to think about edge cases and errors earlier. You pay early with Rust.
- Not Popular - https://madnight.github.io/githut
- Many prefer Go (for webservers and quick

Thinking Non-Functional

- Maintainability (Get it right early - not later)
- Testing (By design)
- Scalability (Low Memory)

You can prototype with the code but it encourages you to have polished code early.

What is Rust?

Based of the Rust Fungus. The logo is a bike ring because most of the developers who started Rust like to ride bikes

used to reading and writing Rust code

- Exercism - Rust Track - Series of exercises to learn rust
- Other resources

```
---
title: Learn with Rustlings
---
stateDiagram-v2
    [*] --> Compile
    Compile --> Fix
    Fix --> Compile
    Compile --> Next
    Next --> Compile
    Next --> [*]
```

Other

Dev Container

Might be flagged by anti-virus software - develop in a container instead 😌

devcontainer.json

```
{
    "name": "Rust",
    "image": "mcr.microsoft.com/devcontainers/rust:latest",
    "customizations": {
        "vscode": {
            "settings": {
                "editor.defaultFormatter": "rust-lang.rust-analyzer",
                "editor.formatOnSave": true,
                "explorer.fileNesting.enabled": true,
                "explorer.fileNesting.expand": false,
                "explorer.fileNesting.patterns": {
                    "Cargo.toml": ".clippy.toml, .rustfmt.toml, Cargo.lock, clippy.toml, cross.toml, rust-toolchain.toml, rustfmt.toml"
                },
                "todo-tree.regex.regex": "(//|#|<!--|;|/\\*|^|^[ \\t]*(-|\\d+.))\\s*($TAGS)|todo!",
            },
            "extensions": [
                "dustypomerleau.rust-syntax",
                "serayuzgur.crates",
                "Gruntfuggly.todo-tree",
                "streetsidesoftware.code-spell-checker"
                "tamasfe.even-better-toml",
                "usernamehw.errorlens",
                "vadimcn.vscode-lldb",
            ]
        }
    }
```

Tools

RustFmt and Clippy

Editions

2021 is the current version - Install and upgrade with https://rustup.rs/

Every three years, 2015, 2018, 2021, 2024. Defaults to 2015 if you don't specify otherwise. Can have breaking changes. You can choose the edition in the Cargo.toml and choose when to upgrade. They provide tooling to automate the upgrade.

https://doc.rust-lang.org/edition-guide/editions/index.html

data,length,capacity
  - Slices: Borrow checker always ensures slices are aways valid at compile time
- Compile time errors
-

```
let mut shopping_list: Vec<&str> = Vec::new(); //Vec<&str> not needed as the compiler can infer it from the next line
shopping_list.push("milk");
```

- Lifetimes
- Return unit (empty tuple)
- Return Result and match //no try catch
- Optional Results

```
pub enum Option<T> {
    None,
    Some(T),
```

Controversy

- Leadership issues
- Trademark Policies
- Choice of Malware writers - 'RustDoor'

Better Programmer